# Unit 3 Classwork — Equations & Inequalities — Answer Key

One/Two-Step · Distributive · Both Sides · Multi-Step · Special Cases · Inequalities · Word Problems

## A. One- & Two-Step Equations

**Problem 1** Score 3

Solve: $4x = 28$

**Answer:** 7

**Problem 2** Score 3

Solve: $3x + 5 = 20$

**Answer:** 5

**Problem 3** Score 3

Solve: $\frac{x}{3} - 2 = 4$

**Answer:** 18

**Problem 4** Score 4

Solve: $-2x + 7 = 1$

**Answer:** 3

## B. Distributive Property & Variables on Both Sides

**Problem 1** Score 3

Solve: $3(2x - 1) = 15$

**Answer:** 3

**Problem 2** Score 4

Solve: $7x - 2 = 3x + 14$

**Answer:** 4

## C. Multi-Step Equations

**Problem 1** Score 4

Solve: $3(x + 2) = 2x + 10$

**Answer:** 4

**Problem 2** Score 4

Solve: $5 - 2x = 3x + 15$

**Answer:** −2

**Problem 3** Score 5

Solve: $4x - 3 = 2(x + 5)$

**Answer:** 6.5

**Problem 4** Score 5

Solve: $\frac{2}{3}(x + 6) - 2 = x + 4$

**Answer:** -6

## D. Special Cases — No Solution or Infinite Solutions

**Problem 1** Score 4

Solve: $2x + 5 = 2x - 1$ — how many solutions?

**Answer:** no solution

**Problem 2** Score 5

Solve: 3(x + 4) = 3x + 12 — how many solutions?

**Answer:** infinitely many solutions

**Problem 3** Score 5

Why does 3(x + 2) = 3x + 10 have no solution?

**A.** Both sides have 3x, so every number works.
**B.** It does have a solution: x = 4/3.
**C.** An equation with parentheses never has a solution.
**D. After you distribute, both sides have 3x, but 6 does not equal 10.**

**Answer:** D

## E. Linear Inequalities

**Problem 1** Score 3

Solve: 3x ≤ 21

**Answer:** x ≤ 7

**Problem 2** Score 3

Solve: 5x − 3 ≥ 12

**Answer:** x ≥ 3

**Problem 3** Score 4

Solve: −2x > 8 (careful!)

**Answer:** x < −4

**Problem 4** Score 5

Solve: −3x + 1 ≤ 10 (careful!)

**Answer:** x ≥ −3

## F. Word Problems

**Problem 1** Score 4

A taxi charges $3 plus $2 per mile. If the total fare was $17, how many miles were driven?

**Answer:** 7

**Problem 2** Score 5

A gym membership costs $25 to join plus $15/month. You can spend at most $100. What's the max number of months?

**Answer:** 5

**Problem 3** Score 5

Plan A: $25 to join a gym plus $15 each month. Plan B: $20 each month with no joining fee. After how many months do the two plans cost the same?

**Answer:** 5